

Title: DATA VORTEX '26 — PHASE 2 ANALYTICAL SQL INSIGHT REPORT

Event & Stage: Aaruush '26 | Round 1 (Phase 2)

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QUESTION 1

E3—Average Engagement by Platform

Challenge: Calculate the average likes, shares, and comments for each platform. Which platform generates the highest average total engagement?

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Primary Finding & Metric Breakdown

YouTube emerges as the highest-performing platform by average total engagement, generating **4,044.55 average total interactions per post**.

Rank	Platform	Avg Likes	Avg Shares	Avg Comments	Avg Total Engagement
1	YouTube	2,528.34	1,011.83	504.38	4,044.55
2	Instagram	2,500.19	1,040.84	499.80	4,040.83
3	Facebook	2,535.65	984.17	506.94	4,026.76
4	Reddit	2,488.95	1,002.23	511.18	4,002.36
5	Unknown	2,467.08	998.61	496.54	3,962.22

6	Twitter	2,437.74	1,005.39	506.13	3,949.26
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Granular Behavioral Patterns

- **Interaction Ratios Across Channels:**
Across all platforms, user behavior follows a strict ratio roughly structured as **62% Likes, 25% Shares, and 13% Comments**. Likes serve as the primary low-friction engagement action across every channel.
- **Channel-Specific Strengths:**
 - **YouTube** leads overall due to high baseline passive approval (2,528.34 likes) paired with strong share performance.
 - **Instagram** holds the highest share rate per post (**1,040.84 shares**), reflecting strong virality mechanisms for visual content.
 - **Reddit** leads in active discussion density with **511.18 comments per post**, outperforming YouTube and Instagram in conversation depth despite ranking lower in total volume.
 - **Twitter** lags behind the top tier across total engagement (**3,949.26**), driven primarily by lower average like counts (2,437.74).

Data Quality Observations

The presence of an **"Unknown"** category ranking 5th (**3,962.22 avg engagement**) highlights an unaddressed upstream tracking issue where platform identifiers were lost prior to ingestion. Because its engagement performance mirrors valid platforms, these rows represent legitimate traffic requiring metadata backfilling rather than arbitrary deletion.

Strategic Recommendations

1. **Format-Driven Optimization:** Capitalize on Instagram's share efficiency for brand awareness campaigns while routing community discussions to Reddit where comment conversion is highest.
2. **Metadata Engineering:** Implement schema validation upstream to catch missing platform tags before records hit production tables, reducing the "Unknown" bucket volume.

QUESTION 2

M1 — Which Locations Generate the Most Engagement? Challenge:
Using both datasets, calculate the total engagement generated by

users from each location. Return the location, number of posts, and total engagement. Rank locations from highest to lowest engagement

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Primary Finding & Top Market Distribution

Los Angeles, USA leads global engagement with 1,838,884 total interactions generated across 459 posts, closely followed by Munich, Germany (1,812,310) and Shanghai, China (1,795,736).

Rank	Location	Post Count	Total Engagement	Avg Engagement / Post
1	Los Angeles, USA	459	1,838,884	4,006.28
2	Munich, Germany	452	1,812,310	4,009.54
3	Shanghai, China	451	1,795,736	3,981.68
4	Barcelona, Spain	439	1,788,641	4,074.35
5	Melbourne, Australia	421	1,717,585	4,079.77
6	Dubai, UAE	421	1,706,751	4,054.04
7	Houston, USA	422	1,669,232	3,955.53

8	Osaka, Japan	398	1,640,653	4,122.24
9	Rio de Janeiro, Brazil	414	1,630,322	3,937.98
10	Mumbai, India	413	1,615,304	3,911.15

Granular Geographic Insights

- **Volume-Driven Ranking Dominance:**

Top-ranked regions secure high position predominantly through raw post publication volume rather than disproportionate virality. The top 3 markets (LA, Munich, Shanghai) maintain the highest volume of posts (>450\$ posts each), showing a linear relationship between total activity and market rank.

- **High-Efficiency Niche Markets:**

Evaluating **average engagement per post** reveals hidden operational efficiency in mid-tier regions:

- **Osaka, Japan (Rank 8):** Delivers the highest per-post interaction rate among major markets at **4,122.24**, outperforming LA (**4,006.28**) despite having 61 fewer posts.
- **Sydney, Australia (Rank 33):** Ranks last in total engagement (**852,028**) due to low post output (**208**), yet achieves an exceptional average per post (**4,096.29**).
- **Barcelona, Spain (Rank 4):** Balances high volume (439 posts) with strong performance (**4,074.35 avg/post**), making it the most balanced European market.

- **Lagging Volume Markets:**

Markets like **Sydney** (208 posts), **Lagos** (223 posts), and **Seoul** (276 posts) sit at the bottom of total engagement rankings strictly due to under-saturated posting activity rather than weak audience interest.

Strategic Recommendations

1. **Volume Expansion Strategy:** Increase creator incentives in high-efficiency, under-saturated regions (e.g., Osaka and Sydney) to capture untapped audience capacity where per-post responsiveness is already elite.
2. **Resource Re-allocation:** Shift paid marketing spend toward secondary markets that demonstrate high per-post conversion rather than over-saturating top tier markets like LA and Munich, where high engagement is diluted across excessive post volume.

QUESTION 3

H4 — Follower to Engagement Anomaly Challenge: Find users who have fewer than 5,000 followers but whose total post engagement places them among the top 10% of all users. These users may represent unusually high performing or suspicious accounts. This requires multiple levels of analysis.

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Primary Anomaly Detection Results

The statistical pipeline successfully isolated **low-follower, top-decile accounts** demonstrating extreme engagement-to-follower asymmetry. The top identified anomaly, **user_uerv85na** (Rome, Italy), achieved **72,361 total interactions across 16 posts** despite possessing only **1,824 followers**.

Rank	User ID	Location	Follow er Count	Post Cou nt	Total User Engageme nt	Engageme nt Decile
1	user_uerv85na	Rome, Italy	1,824	16	72,361	1
2	user_hdas0ia u	Rio de Janeiro, Brazil	1,620	16	64,007	1
3	user_n0ok02r t	Dubai, UAE	2,531	18	62,920	1
4	user_fgjkkrie	Lyon, France	2,211	14	62,690	1

5	user_6wra58f7	Johannesburg, South Africa	4,953	14	57,476	1
6	user_5oe5t3js	London, UK	3,151	13	55,241	1
7	user_rr1uzkql	Milan, Italy	1,069	14	54,632	1
8	user_67hyf45u	Vancouver, Canada	898	11	52,098	1
9	user_zwtxsrs	Houston, USA	1,009	13	50,584	1
10	user_knjtsbf3	Toronto, Canada	1,298	11	48,959	1

Granular Behavioral & Behavioral Risk Analysis

- **Extreme Reach Multipliers:**
 1. **user_67hyf45u (Vancouver, Canada):** Demonstrates the most severe ratio anomaly in the dataset. With only **898 followers**, this account generated **52,098 total interactions across 11 posts**, averaging **4,736 interactions per post**. This equates to a reach multiplier exceeding **527% of their follower base per post**.
 2. **user_uerv85na (Rome, Italy):** Averaged **4,522.5 interactions per post** with 1,824 followers (~248% reach efficiency per post).
- **Dual-Scenario Classification Framework:**
 1. **Pattern A: Organic Micro-Virality (High Potential):**
Accounts like **user_6wra58f7** (4,953 followers, 57,476 engagement) operate closer to organic growth thresholds. High share volumes on specific posts can push small accounts into off-network recommendation loops (e.g., algorithmic discovery feeds).
 2. **Pattern B: Coordinated Inauthentic Behavior / Bot Networks (High Risk):**
Accounts like **user_67hyf45u** (<900K followers, >52K engagement) exhibit

activity inconsistent with organic audience distribution. Generating 4,700+ interactions per post without an established follower base strongly signals external engagement farms, automated click-bots, or cross-platform comment rings.

Strategic & Operational Recommendations

1. **Automated Audit Pipeline:** Route all 16 isolated user IDs directly to the Trust & Safety audit table for secondary payload analysis (evaluating timestamp velocity and share-to-like ratios).
2. **Growth Program Recruitment:** For accounts verified as organic (non-bot traffic), enroll them in platform creator accelerator programs to monetize hyper-viral micro-creators before they churn to competing platforms.